Histology Syllabus

MED105 • Anne Robinson

Life Sciences • Medicine

1. Learning Objectives & Taxonomy

ID	Learning Objective	Bloom's Level
LO1	Demonstrate proficiency in operating microscopes and applying fundamental fixation, sectioning, and staining techniques for tissue sample preparation.	Applying
LO2	Differentiate between the structural and functional characteristics of various epithelial, connective, muscle, and nervous tissues.	Analyzing
LO3	Examine the microscopic architecture of organ systems including integumentary, cardiovascular, lymphoid, respiratory, urinary, and endocrine systems.	Analyzing
LO4	Evaluate the histological organization of the digestive tract and its accessory glands in relation to metabolic and physiological processes.	Evaluating
LO5	Correlate normal microscopic anatomy and cellular morphology with clinical pathology across diverse organ systems.	Evaluating

Overall Course Cognitive Depth: Analyzing, Evaluating, Understanding

2. Weekly Instructional Plan

Wk	Subject and Activities	Outcomes
1	**Introduction to Histology & Microscopy** Face to Face Lectures, Syllabus	LO1
2	**Epithelial Tissue** Face to Face Lectures • Lab Sessions	LO2
3	**Connective Tissue Proper** Discussion Forums, Face to Face Lectures • Lab Sessions • Quizzes	LO2
4	**Specialized Connective Tissues: Cartilage and Bone** Lab Report, Face to Face Lectures • Lab Sessions • Case Study	LO2
5	**Blood and Hematopoiesis** Face to Face Lectures • Lab Sessions • Simulations	LO2
6	**Muscle Tissue** Face to Face Lectures • Think-Pair-Share • Quizzes	LO2
7	**Nervous Tissue** Article Analysis, Face to Face Lectures • Lab Sessions • Midterm Exam	LO2
8	**Integumentary System** Video Lectures • Case Study • Assignments / Homework	LO3
9	**Urinary System** Tutorial Sessions • Group Exercises • Quizzes	LO3, LO5
10	**Lymphoid System** Face to Face Lectures • Lab Sessions	LO3
11	**Digestive System I (Oral Cavity to Stomach)** Face to Face Lectures • Lab Sessions • Quizzes	LO4
12	**Digestive System II (Glands & Accessory Organs)** Face to Face Lectures • Lab Sessions • Case Study	LO4
13	**Respiratory System & Urinary System** Face to Face Lectures • Lab Sessions • Lab Report	LO3, LO5
14	**Endocrine System** Peer Learning, Face to Face Lectures • Lab Sessions	LO3, LO5
15	**Reproductive System & Special Senses** Practice Exams, Face to Face Lectures • Group Discussions, Lab Sessions	LO3, LO5
16	**Final Exam** Final Exam	—

3. Skill Acquisitions by Category

Domain	Specific Skill	Related Wk
Hard	Microscopy and Tissue Preparation	1
	Epithelial and Connective Tissue Identification	2, 3
	Skeletal and Musculoskeletal Histology	4
	Blood Smear Analysis and Hematopoiesis	5
	Muscle Tissue Differentiation	6
	Nervous Tissue and Glial Cell Identification	7
	Integumentary System Microscopic Analysis	8
	Cardiovascular and Urinary Architecture Evaluation	9, 13
	Lymphoid and Immune Organ Examination	10
	Gastrointestinal Tract Histological Assessment	11, 12
	Endocrine Gland and Special Senses Evaluation	14, 15
Soft	Collaborative Microscopy Lab Work	1, 4, 11, 15
Meta	Clinical Diagnostic Reasoning	7, 12, 16

4. Proposed Micro-Credential Framework

Credential Name	Dimensions	Depth	Align.	Mapping
Clinical Histopathology Specialist	Application	Proficiency	W1-16	#MicroscopySkills #DiagnosticMicroscopy #TissueBiology
Organ Systems Micro-Anatomy Analyst	Integration	Mastery	W8-15	#OrganHistology #DiagnosticPathology #MedicalPhysiology

5. Assessment & Mapping

Activity Name	Week	Weight	Bloom's	Outcomes
Quizzes	3	9%	Analyzing	LO2
Lab Report	4	14%	Evaluating	LO2, LO3
Quizzes	6	9%	Analyzing	LO2
Midterm Exam	7	23%	Analyzing	LO1, LO2, LO3
Assignments	8	9%	Applying	LO3
Quizzes	9	9%	Analyzing	LO3
Quizzes	11	9%	Analyzing	LO4
Final Exam	16	18%	Evaluating	LO1, LO2, LO3, LO4, LO5